Agenda

Strategy Overview

- Q-Learning (Reinforcement Learning)
- Investment Securities
- Competitive Edge

The Model

- Empirical Exploration
- Implications
- Model Construction
- Implementations

Backtesting

- Assumptions
- Model Stability
- Return and Risk
- Cumulative Returns
- Sharpe Ratio

Risk Management

- By Portfolio Structure
- By Process

Performance

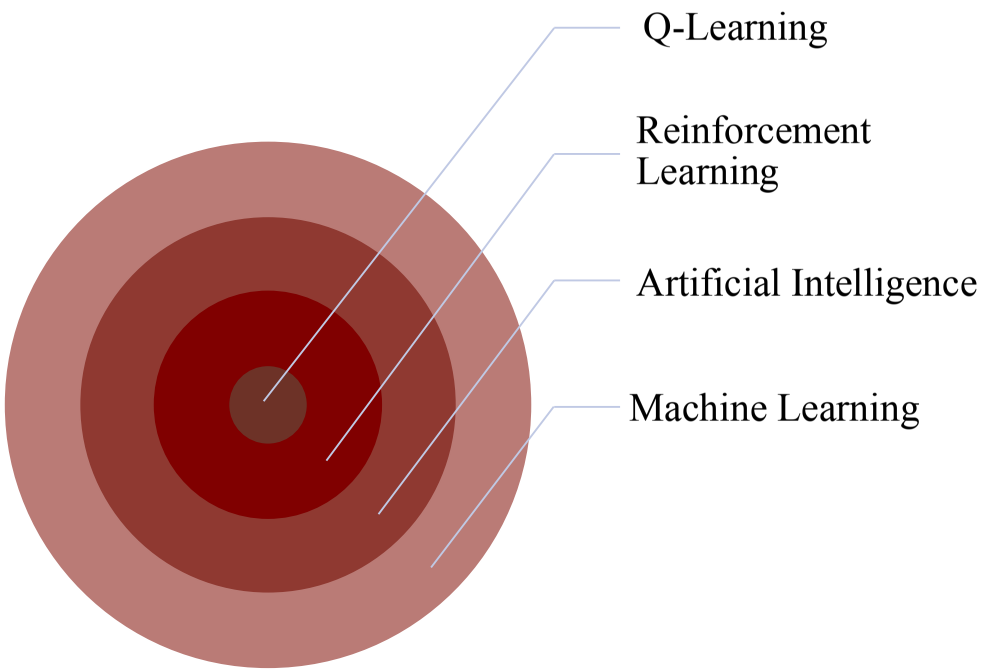
- Portfolio Building
- Performance to Date

About

- The Managers

Strategy Overview – Q Learning

- A branch of Reinforcement Learning
- Broadly applied to sequential decision making problems.
- Used to find optimal action-selection trading policy
- Dynamically process feedbacks from the previous strategy data.

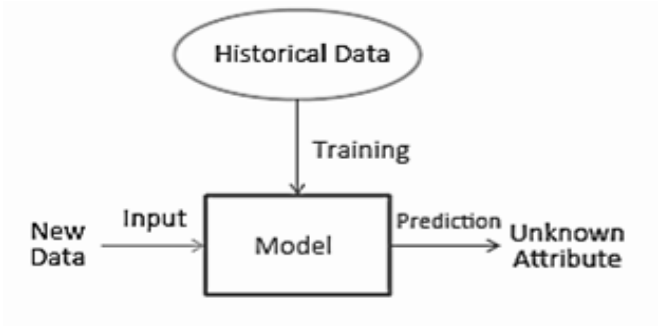


Investment Universe

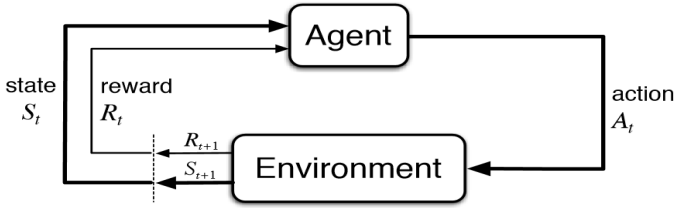
Investment Securities:
• Foreign Exchange Currencies publicly traded on international exchange • The most liquid ETFs publicly traded on exchange • All securities are highly liquid and actively traded, positions can be enter and exit without any barrier

The Strategy Takes:
• Long and short positions in Foreign Exchange Currencies • Long and short positions in ETFs • High frequency trading (120 seconds interval)

Competitive Edge Q&A



Classical Machine Learning Process



Q-Learning Process

Q: Is there any proven success of Reinforcement Learning?

A: Yes, AlphaGo is based on Reinforcement Learning and it has just beat the best human Go player.

Q: Why should this work in trading?

A: Trading strategy is a sequential decision making process that exactly fits the mechanism.

Q: What's the difference of Q-Learning, as a branch of reinforcement learning, compared to classical machine learning?

A: It is more than classical machine learning prediction algorithms. It includes the feedback into future training and future actions. See comparisons on the left.

Q: What is the biggest advantage of Q-Learning?

A: With feedback, less overfits will occur compared to other machine learning algorithms.

Empirical Exploration

Currently, when applying Reinforcement learning and Q-learning to quantitative strategies, researchers tried to get better results by **maximizing value functions**.

Pros

Customized value function for each dataset yields the best fit

Cons

- The value function for each dataset varies dramatically
- Each value function only works for one dataset
- Model is not reusable
- Very time and money consuming
- Requires extremely powerful server to find the value function

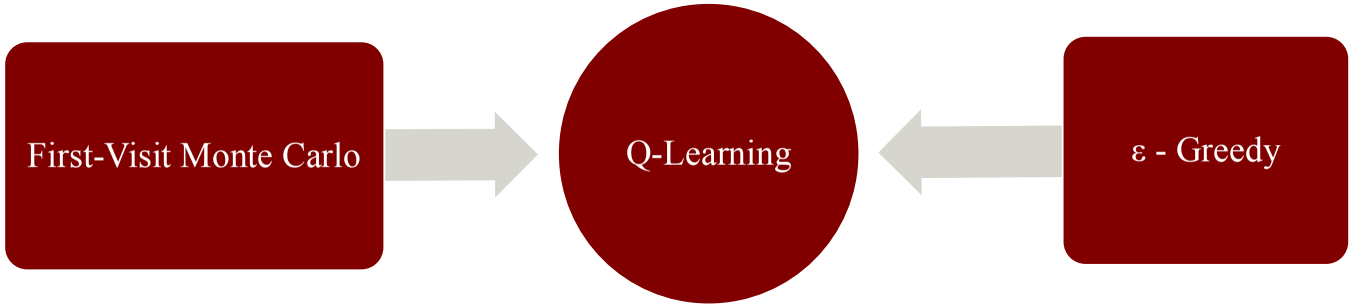
Clearly, in practice, the benefit can barely cover the cost.

Implications of Empirical Exploration

Then, we ask ourselves the defining question:

How can we develop a reusable model that is applicable to all dataset?

After careful research, we decided to replace value function by the following two methodologies:



And then, we have our model……

Model Construction

Define Q(s,a) as the measure of **goodness of fit** at **state-action** combination (S,A):

$$Q(s_t,a_t) \leftarrow \underbrace{Q(s_t,a_t)}_{old-value} + \underbrace{\alpha_t}_{learning-rate} \cdot (\overbrace{\underbrace{r_{t+1}}_{reward} + \underbrace{\gamma}_{discount-factor} \cdot \max_a Q(s_{t+1},a)}^{learned-value} - \underbrace{Q(s_t,a_t)}_{old-value})$$

s_t is the state, a_t is the action, α is the learning rate, r is the reward and γ is the discount factor.

$Q(s_t, a_t)$ is derived from old-value, learning-rate multiplies the result of learned-value minus old value (see formula above). This process is named **temporal difference (TD).**

First-Visit Monte Carlo is also implemented to evaluate $Q(s_t, a_t)$ as we average the returns for the first time state St is visited in an episode. The **ε-greedy strategy** is applied to weighted-average a series of initial empirical returns which guarantees the First-Visit Monte Carlo algorithm and profit to **asymptotically converge, and more importantly,** guarantees **the improvement policy.** This means, the employment of First-Visit Monte Carlo and **ε-greedy successfully replaced the value function.** More theoretical details are explained in the technical paper.

Our Implementation

The Model

- A Q-Learning Model for each financial product
- Each model is individually tuned and trained
- On average, 2,000 data points are trained under each model to ensure convergency and stability
- First-Visit Monte Carlo and ε – greedy methodologies were employed

Traded Financial Products

- Buy and short the selected 6 liquid foreign exchange currencies
- Buy and short the selected 6 highly liquid ETF
- Each product's return trend is individually learned by a model
- We review active position every 10 minutes
- Each position must hold at least 10 minutes (5 period, 2 minutes per period)

Backtesting: Assumptions

Data

- 5/1/2017 – 5/5/2017
- High Frequency data, 2 minutes per data point
- Data source: From Bloomberg
- Volume Check: Ensures actual traded volume being lower than our simulated volume

Costs, Expenses and Fees

- We are market makers and the **transaction cost will be rebated from exchange**. However, a profit and loss analysis with fixed transaction cost $5/transaction will be shown at the end for other party's reference (see technical paper).
- We do high frequency trading, so the **borrowing interest** is very little to consider.
- There is no **Management or Incentive Fees** because we are still in our Beta stage.

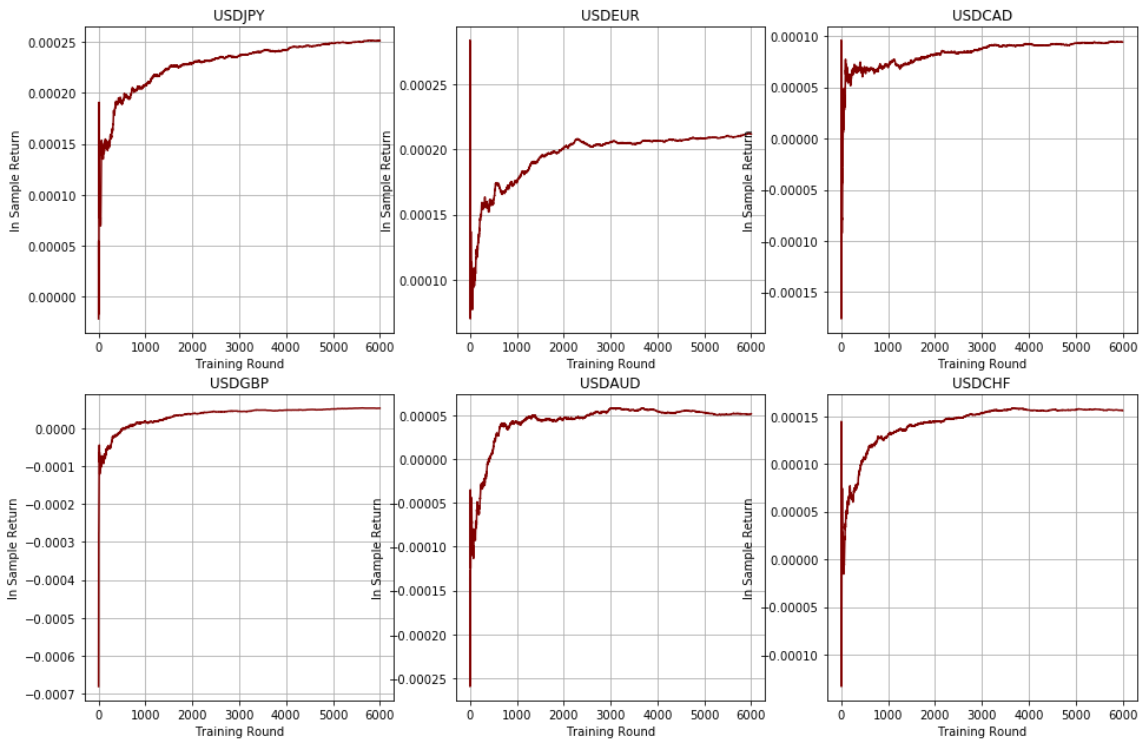
Others

- The decision (buy/short) not only depends on price, but also the short term and long term behavior Sharpe Ratios that describe status. Term length are parameters to tune under each model.
- At the beginning of training period, we assume we hold only long positions until we get the first Sharpe ratio we need. (usually takes 20 training data points)

Portfolio

- Three testing portfolios. Portfolio A: 6 foreign exchange currencies. Portfolio B: 6 ETFs. Portfolio C: Combination of Portfolio A and B.
- Each product is allocated $100,000 capital in portfolio A and B. $50,000 for each product in portfolio C.
- $1 Million capital is allocated to portfolio A and $1 Million capital to portfolio B. $1 Million capital allocated to portfolio C

Model Stability: Foreign Exchange



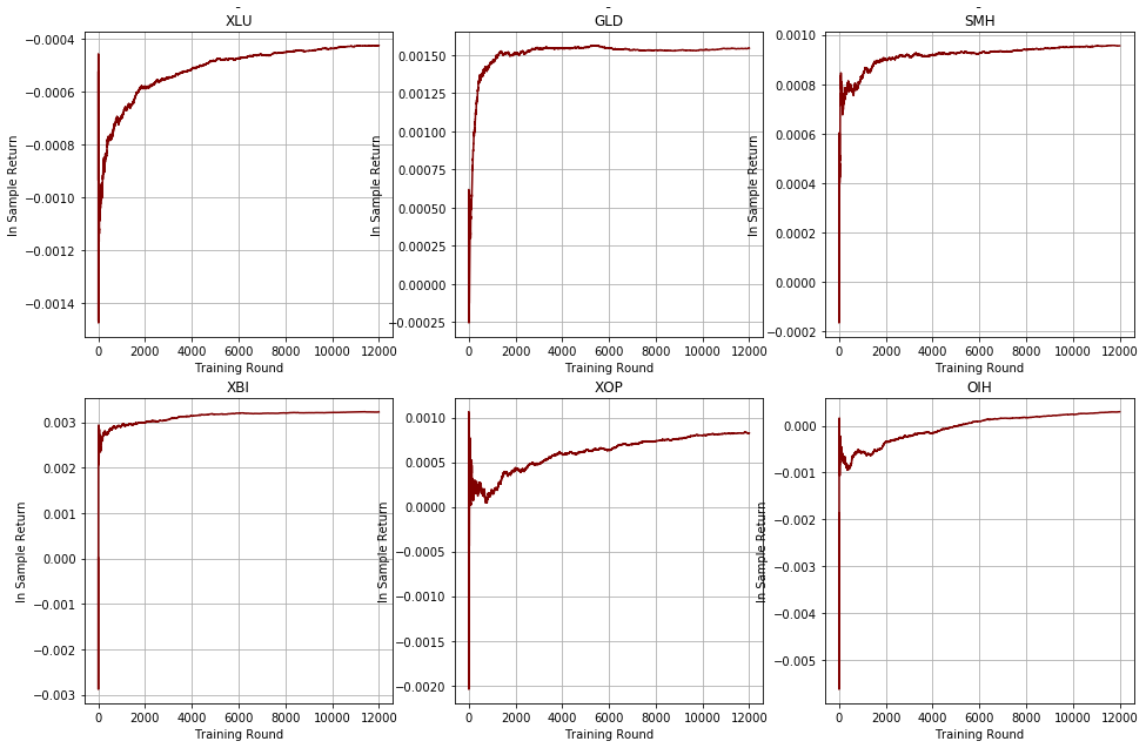
All models for Foreign Exchange products are well trained to have in-sample returns **converged**.

We conclude:

Models are stable and ready to use.

Now let's see the models for ETFs

Model Stability: ETF



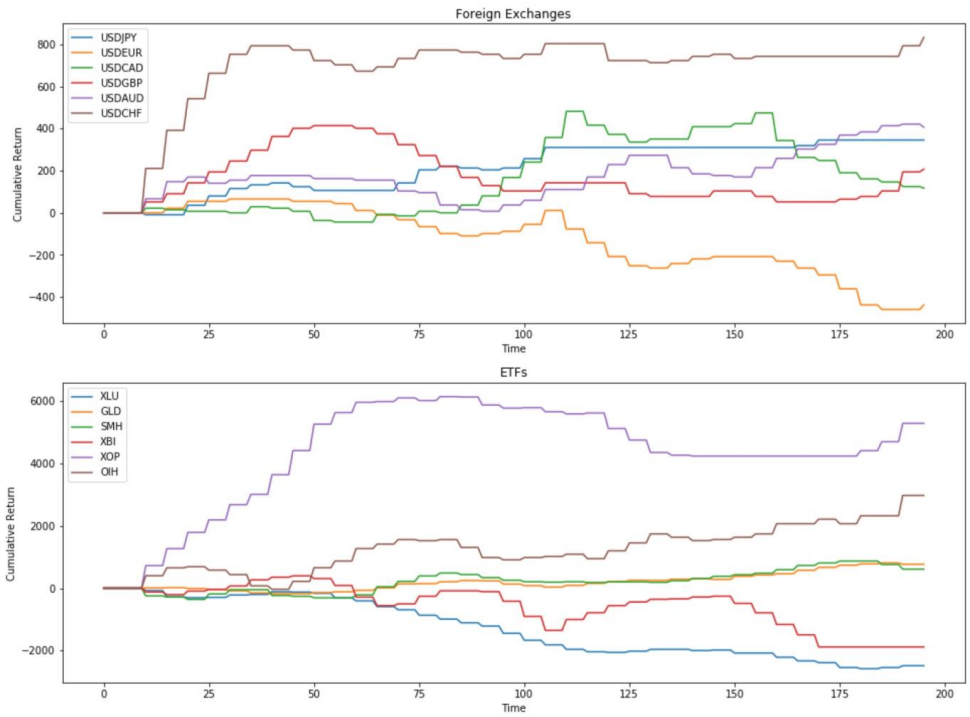
All models for ETF products are well trained to have in-sample returns **converged**.

We conclude:

Models are stable and ready to use.

Now, we are going to trade.

Results: Individual Cumulative Returns



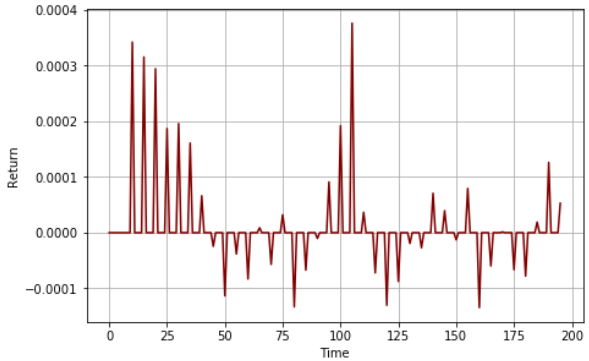
* Each time period is 2 minutes

What can we learn from the returns?

- Foreign Exchanges: Although the models are all well-tuned, we can still see discrepancies in returns. For example, while USDJPY is generating profit, USDCHF is losing money.
- ETFs: The same earnings discrepancy occurs on ETF products as well.
- Hence, we know, investing in single product is very risky.
- To diversify the risk, we would like to build three portfolios: A: 6 Foreign Exchanges, B: 6 ETFs and C: the combination of A and B.

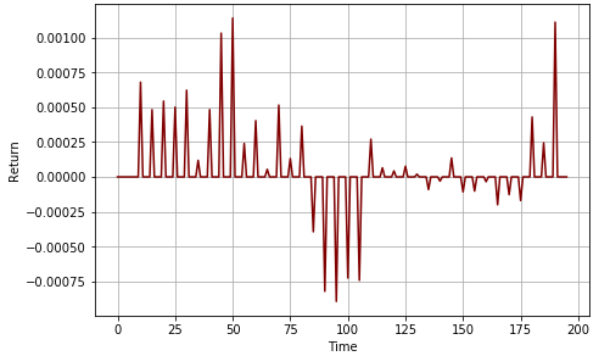
Results: Portfolio Returns and Risk

Return of Portfolio A



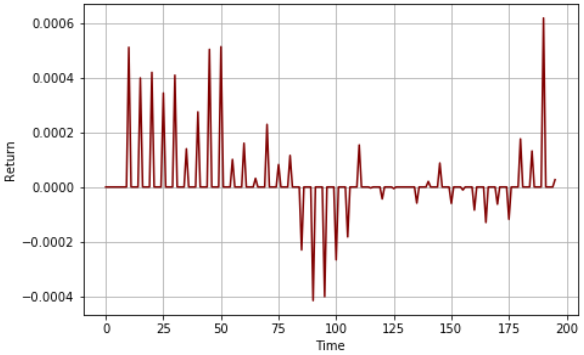
Risk Measure	Value
VaR 0.1%	$180.91
VaR 1%	$134.33
VaR 5%	$92.78
VaR 10%	$70.62
Standard Deviation	6.098e-5

Return of Portfolio B



Risk Measure	Value
VaR 0.1%	$640.12
VaR 1%	$475.25
VaR 5%	$328.17
VaR 10%	$249.75
Standard Deviation	0.000216

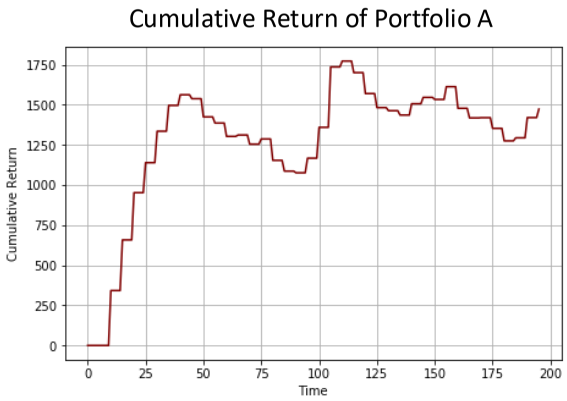
Return of Portfolio C

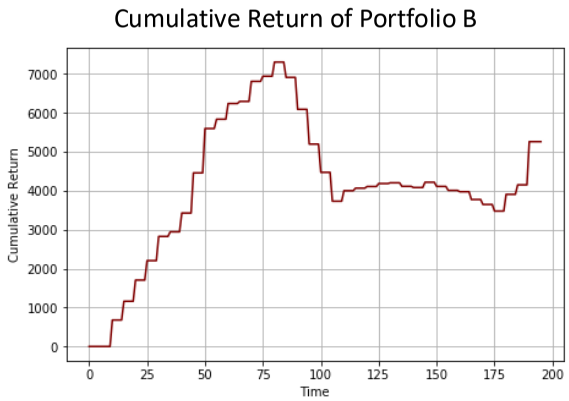


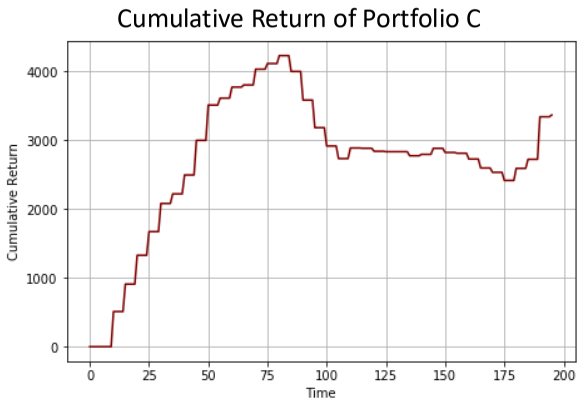
Risk Measure	Value
VaR 0.1%	$335.15
VaR 1%	$248.06
VaR 5%	$170.36
VaR 10%	$128.94
Standard Deviation	0.000114

* Each time period is 2 minutes

Results: Portfolio Cumulative Returns







Inherent the risk parameters from last slides, we have the following comparisons:

Return	**Low**
Risk (Std)	Low

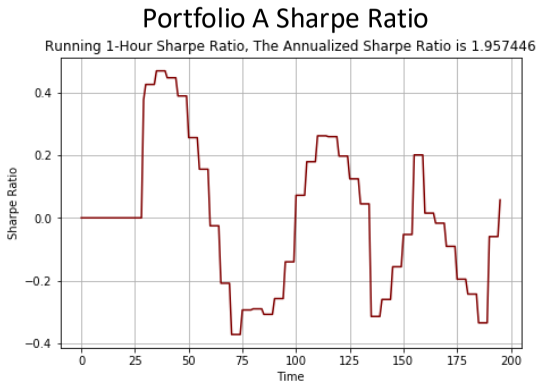
Return	**High**
Risk (Std)	High

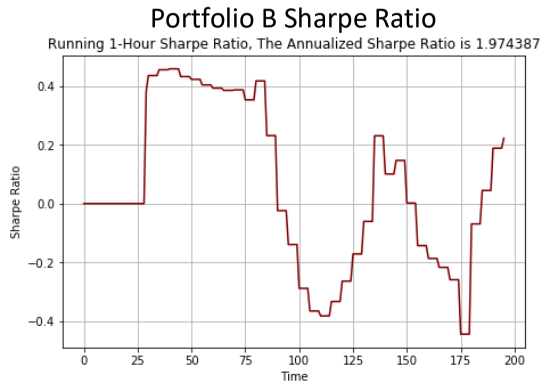
Return	**Medium**
Risk (Std)	Medium

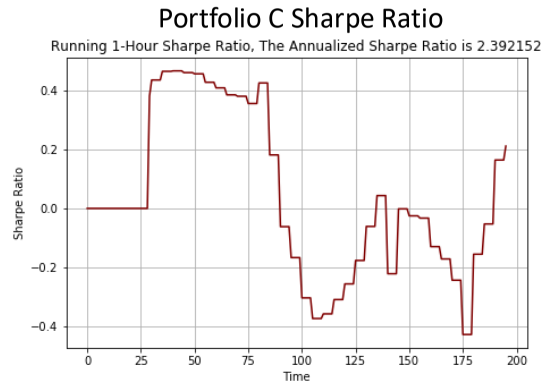
Let's review the Sharpe ratio for further evaluation.

* Each time period is 2 minutes

Results: Sharpe Ratio







We found, although portfolio C has the medium risk and the medium return among the three portfolios, it has the highest Sharpe ratio, which means, it has the most attractive risk adjusted return.

We also found, although each model for each product is well trained, it does not guarantee positive returns. Hedge multiple products against each other is necessary to create an attractive portfolio.

* Each time period is 2 minutes

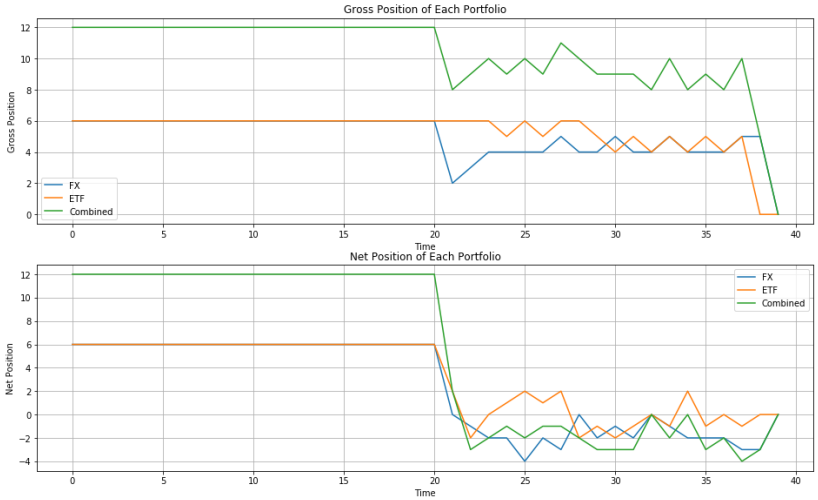
Risk Management: By Portfolio Structure

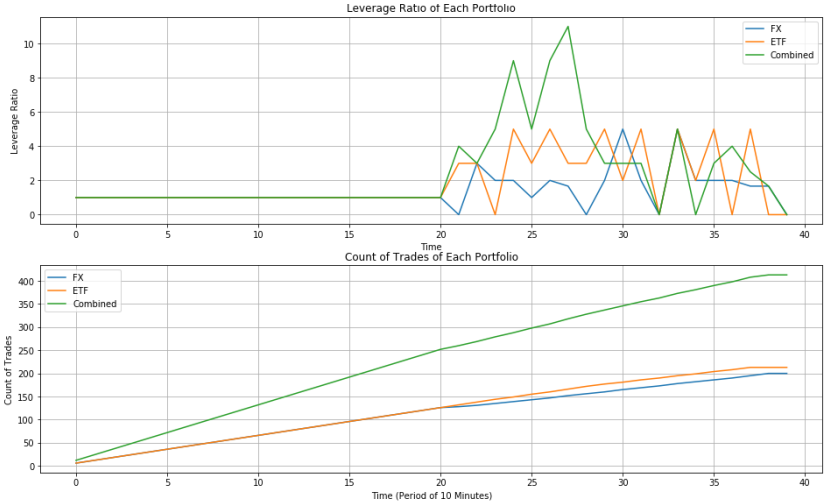
Portfolio A	Portfolio B	Portfolio C
• Diversified portfolio of long and short positions • Limits on 60% of capital to be invested in the foreign exchange market sector • Limits on 10% of capital to be invested in each asset	• Diversified portfolio of long and short positions • Limits on 60% of capital to be invested in the ETF market sector • Limits on 10% of capital to be invested in each asset	• Diversified portfolio of long and short positions • Combination of two sectors (Foreign Exchange and ETF) • Limits of 30% of capital to be invested in each sector • Limits of 5% of capital to be invested in each asset

Risk Management: By Process

- From the model:
  - We tuned the model to have the optimal learning rate and discount factor to eliminate risk to the largest extent
- Regular review of each position
  - Every 10 minutes, or upon significant daily events
- Maintain/exit decision
  - Re-balancing every 10 minutes
- Stop loss
  - if 10-minute loss from position > 1% of individual position price
- Re-train the model every day before market open to ensure model stability, ensure the in-sample returns converge.
- Real-time monitoring of price moves and news items

Launch and Portfolio Building





- Launched on 5/8/2017 8:30am to 3:00pm
- Each position is limited to $100,000 for A and B, $50,000 for C.
- 1% stop loss
- 1 Million capital is allocated to portfolio A and B respectively, 2 Million capital is allocated to portfolio C
- Less than 60% of portfolio capital is utilized for risk control, as our company is still in Beta stage.

* Each time period is 10 minutes

- **Long positions will be held until the model collect enough information to make rational decisions, which causes the horizon in the plot.**
- Three portfolio options will be available, with various returns and volatilities.
- Due to the special feedback characteristic of our model, the model needs to be trained every morning before trading begins.
- Each model training will consume about 3 hours.